WORKED SOLUTION

STEP III 1997, Question 1

Pure • Exponential Power Series

STEP III 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q1

The key idea

Multiplying $e^x = \sum x^n/n!$ by a polynomial shifts the series along. The coefficient of x^n is then a combination of $1/n!$, $1/(n-1)!$, $1/(n-2)!, \ldots$, and over the common denominator $n!$ the numerator is a polynomial in n. Putting $x = 1$ turns the identity of power series into an identity for a numerical series. So the whole question is about choosing the polynomial that produces the numerator we want. The *falling factorials* n, $n(n-1)$, $n(n-1)(n-2)$ are the natural building blocks, because $n(n-1)\cdots(n-k+1)/n! = 1/(n-k)!$.

Solution

Throughout, $e^x = \sum_{n\geq 0} x^n/n!$ converges absolutely for every real x. Multiplying it by a polynomial gives a finite sum of such series, so we may add the series term by term and then set $x = 1$. We use the convention $1/m! = 0$ for negative integers m. This lets every sum below start at $n = 0$.

Part (i)

We have

$$(x^2+5x+4)\,e^x = \sum_{n\geq 0}\frac{x^{n+2}}{n!} + 5\sum_{n\geq 0}\frac{x^{n+1}}{n!} + 4\sum_{n\geq 0}\frac{x^n}{n!}.$$

To collect the coefficient of x^n, reindex the first two sums. The coefficient is

$$\begin{aligned}\frac{1}{(n-2)!} + \frac{5}{(n-1)!} + \frac{4}{n!} &= \frac{n(n-1)+5n+4}{n!}\\ &= \frac{n^2+4n+4}{n!} = \frac{(n+2)^2}{n!}.\end{aligned}$$

This holds for every $n \geq 0$, including $n = 0$ and $n = 1$, where the convention $1/m! = 0$ removes the missing terms. (For $n = 0$ the coefficient is $4 = 2^2/0!$; for $n = 1$ it is

$5 + 4 = 9 = 3^2/1!$.) Hence

$$(x^2 + 5x + 4)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+2)^2}{n!}\,x^n,$$

and putting $x = 1$ gives

$$10\,\mathrm{e} = \frac{2^2}{0!} + \frac{3^2}{1!} + \frac{4^2}{2!} + \frac{5^2}{3!} + \cdots = 4 + \frac{3^2}{1!} + \frac{4^2}{2!} + \cdots,$$

as required.

Part (ii)

Now the general term is $(n+1)^2/n!$, so we want a quadratic $x^2 + bx + c$ whose series has the coefficient

$$\frac{n(n-1) + bn + c}{n!} = \frac{(n+1)^2}{n!}.$$

Ask yourself How do we find b and c without guessing?

Write the target numerator in falling factorials: $n^2 + 2n + 1 = n(n-1) + 3n + 1$. Matching gives $b = 3$ and $c = 1$. So, by exactly the argument of part (i),

$$(x^2 + 3x + 1)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+1)^2}{n!}\,x^n,$$

and putting $x = 1$ gives

$$(1 + 3 + 1)\,\mathrm{e} = 5\,\mathrm{e} = 1 + \frac{2^2}{1!} + \frac{3^2}{2!} + \frac{4^2}{3!} + \cdots.$$

Part (iii)

The general term is $(n+1)^3/n!$, so we now need a cubic. The coefficient of x^n in $(ax^3 + bx^2 + cx + d)\,\mathrm{e}^x$ is

$$\frac{a}{(n-3)!} + \frac{b}{(n-2)!} + \frac{c}{(n-1)!} + \frac{d}{n!} = \frac{a\,n(n-1)(n-2) + b\,n(n-1) + c\,n + d}{n!}.$$

We expand the target in falling factorials. Since $n^3 = n(n-1)(n-2) + 3n(n-1) + n$ and $n^2 = n(n-1) + n$,

$$\begin{aligned}(n+1)^3 &= n^3 + 3n^2 + 3n + 1\\ &= [n(n-1)(n-2) + 3n(n-1) + n] + 3[n(n-1) + n] + 3n + 1\\ &= n(n-1)(n-2) + 6n(n-1) + 7n + 1.\end{aligned}$$

(A quick check at $n = 1$: both sides equal 8. At $n = 2$: $27 = 0 + 12 + 14 + 1$.) So $a = 1$, $b = 6$, $c = 7$, $d = 1$ and

$$(x^3 + 6x^2 + 7x + 1)\,\mathrm{e}^x = \sum_{n\geq 0} \frac{(n+1)^3}{n!}\,x^n.$$

Putting $x = 1$,

$$1 + \frac{2^3}{1!} + \frac{3^3}{2!} + \frac{4^3}{3!} + \cdots = (1 + 6 + 7 + 1)\,\mathrm{e} = 15\,\mathrm{e}.$$

Ask yourself Why can't we just square or cube the series term by term?

Because $\sum (n+1)^3/n!$ is not built from $\sum (n+1)/n!$ in any simple way. The sum is linear in the numerator, so what we can do is split the numerator into pieces $n(n-1)\cdots(n-k+1)$, each of which cancels against $n!$ to give a shifted copy of the series for e. Each such piece contributes exactly e at $x = 1$. The answer is therefore e times the sum of the falling-factorial coefficients, $1 + 6 + 7 + 1 = 15$.

Checks

- Partial sums up to $n = 60$ give $27.1828\ldots$, $13.5914\ldots$ and $40.7742\ldots$, agreeing with 10e, 5e and 15e to machine precision.
- The coefficients in the three numerator identities were checked exactly for $n = 0, \ldots, 39$.
- Consistency: part (ii)'s series is $\sum (n+1)^2/n!$ and part (i)'s is $\sum (n+2)^2/n!$. The difference $\sum (2n+3)/n! = 2\sum n/n! + 3\sum 1/n! = 2\mathrm{e} + 3\mathrm{e} = 5\mathrm{e}$ is indeed $10\mathrm{e} - 5\mathrm{e}$.

Common slips

- Forgetting that the first two terms ($n = 0, 1$) need separate attention when sums are reindexed. The convention $1/m! = 0$ for $m < 0$ handles them, but you should check them.
- Matching $(n+1)^3$ against powers n^3, n^2, n instead of falling factorials, and then picking the wrong cubic.

Beyond STEP

The sums $\sum_{n \geq 0} n^k/n!$ are $B_k\,\mathrm{e}$, where B_k is the kth Bell number (the number of ways to partition a set of k objects): $B_0, \ldots, B_4 = 1, 1, 2, 5, 15$. This is Dobiński's formula. Indeed, writing $m = n + 1$, part (iii) is $\sum_{m \geq 1} m^4/m! = B_4\,\mathrm{e} = 15\,\mathrm{e}$, and the coefficients $1, 6, 7, 1$ are the Stirling numbers of the second kind $1, 7, 6, 1$ in reverse order. Rearranging and combining power series term by term is justified by absolute convergence, a theme of first-year analysis. See Abbott, *Understanding Analysis*, for the theory, and Siklos, *Advanced Problems in Mathematics*, for more questions of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.